STA2002 Probability and Statistics II
Assignment 4
Due date: **12 November 2023 11:59pm**
Please submit your written answers as a pdf file.

Please work on the assignment by yourself only. We follow a strict rule on academic honesty.

1. Let X be the weights of newborn babies in pounds, and denote m to be its median. For a sample of 14, we have:

   5.2 11.8 7.1 14.4 6.3 7.5 11.1 5.5 9.8 12.7 10.5 10.9 8.7 9.2

   (a) Give point estimates of $\pi_{0.25}, \pi_{0.35}, m, \pi_{0.75}$.

   $$\widehat{\pi}_{0.25} = x_{(3)} + 0.75(x_{(4)} - x_{(3)}) = 6.3 + 0.75(7.1 - 6.3) = 6.9.$$
   $$\widehat{\pi}_{0.35} = x_{(5)} + 0.25(x_{(6)} - x_{(5)}) = 7.5 + 0.25(8.7 - 7.5) = 7.8.$$
   $$m = \widehat{\pi}_{0.5} = x_{(7)} + 0.5(x_{(8)} - x_{(7)}) = 9.2 + 0.5(9.8 - 9.2) = 9.5.$$
   $$\widehat{\pi}_{0.75} = x_{(11)} + 0.25(x_{(12)} - x_{(11)}) = 11.1 + 0.25(11.8 - 11.1) = 11.275.$$

   (b) Find the following confidence intervals and state the associated confidence coefficient:
   i. $(X_{(1)}, X_{(5)})$, a confidence interval for $\pi_{0.25}$.
   ii. $(X_{(3)}, X_{(8)})$, a confidence interval for $\pi_{0.35}$.
   iii. $(X_{(5)}, X_{(10)})$, a confidence interval for $\pi_{0.5}$.
   iv. $(X_{(8)}, X_{(13)})$, a confidence interval for $\pi_{0.75}$.

   Part (i). The interval is $(x_{(1)} = 5.2, x_{(5)} = 7.5)$. Confidence coefficient is

   $$\sum_{k=1}^{4} \binom{n}{k} (0.25)^k (0.75)^{n-k} = 0.7415 - 0.0178 = 0.7237.$$

   Part (ii). The interval is $(x_{(3)} = 6.3, x_{(8)} = 9.8)$. Confidence coefficient is

   $$\sum_{k=3}^{7} \binom{n}{k} (0.35)^k (0.65)^{n-k} = 0.9247 - 0.0839 = 0.8408.$$

   Part (iii). The interval is $(x_{(5)} = 7.5, x_{(10)} = 10.5)$. Confidence coefficient is

   $$\sum_{k=5}^{9} \binom{n}{k} (0.5)^k (0.5)^{n-k} = 0.9102 - 0.0898 = 0.8204.$$

   Part (iv). The interval is $(x_{(8)} = 9.8, x_{(13)} = 12.7)$. Confidence coefficient is

   $$\sum_{k=8}^{12} \binom{n}{k} (0.75)^k (0.25)^{n-k} = \sum_{k=2}^{6} \binom{n}{k} (0.75)^{n-k} (0.25)^k = 0.9617 - 0.1010 = 0.8607.$$

(c) Now, we would like to test $H_0 : m = 8$ against $H_1 : m < 8$ at significance level $\alpha = 0.05$.

i. Use the sign test to test the hypothesis.
Let n^- be the number of data that is < 8. Then $n^- = 5$. The p-value is $P(N^- \geq 5; H_0) = 1 - P(Bin(14, 0.5) < 5) = 1 - 0.0898 = 0.9102 > 0.05$, so we fail to reject H_0 at 5%.

ii. Use R to conduct the Wilcoxon sign-rank test to test the hypothesis (hint: you can use the wilcox.test() function and set the 'exact' parameter to FALSE).

```
> data1
 [1]  5.2 11.8  7.1 14.4  6.3  7.5 11.1  5.5  9.8 12.7 10.5 10.9  8.7  9.2
> wilcox.test(data1,mu = 8,alternative=c('less') ,exact=FALSE)

        Wilcoxon signed rank test with continuity correction

data:  data1
V = 79.5, p-value = 0.9579
alternative hypothesis: true location is less than 8
```

Supplementary:
The Wilcoxon test statistic is $w = 54$. The p-value is

$$P(W \leq 54) = P(W < 55) = \Phi\left(\frac{55}{\sqrt{(14)(15)(29)/6}}\right) = \Phi(1.7264) = 0.9579 > 0.05,$$

so we fail to reject H_0 at 5%.

iii. Use the t-test to test the hypothesis, assuming the underlying distribution is symmetric.
The test statistic is

$$t = \frac{\overline{x} - m_0}{s/\sqrt{n}} = 1.802 > -t_{0.05}(13) = -1.771,$$

so we fail to reject H_0 at 5%.

2. Let X be the number of blueberries on a muffin made by Michael, and denote by m to be its median. Charles suspects that Michael has been adding more blueberries than usual to his muffins. For some reason, he has kept meticulous records. The last 15 muffins Charles bought contained the following numbers of blueberries:

8 10 7 12 15 8 10 10 11 12 9 13 6 11 6

Charles would like to test $H_0 : m = 9$ against $H_1 : m > 9$ at significance level $\alpha = 0.1$.

(a) Use the sign test to test the hypothesis.
Let n^+ be the number of data that is > 9. Then $n^+ = 9$. The p-value is

$$P(N^+ \geq 9; H_0) = 1 - P(Bin(15, 0.5) < 9) = 1 - 0.6964 = 0.3036 > 0.1,$$

so we fail to reject H_0 at 10%.

(b) Use R to conduct the Wilcoxon sign-rank test to test the hypothesis (hint: set the 'exact' parameter to FALSE).

```
> data2
 [1]  8 10  7 12 15  8 10 10 11 12  9 13  6 11  6
> wilcox.test(data2,mu = 9,alternative=c('greater') ,exact=FALSE)

        Wilcoxon signed rank test with continuity correction

data:  data2
V = 71, p-value = 0.1272
alternative hypothesis: true location is greater than 9
```

Supplementary:
The Wilcoxon test statistic is $w = 37$. The effective sample size is $n = 14$. The p-value is

$$P(W \geq 37) = 1 - P(W \leq 35) = 1 - \Phi\left(\frac{36}{\sqrt{(14)(15)(29)/6}}\right) = 1 - \Phi(1.1300) = 0.1292 > 0.1,$$

so we fail to reject H_0 at 10%.

(c) Use the t-test to test the hypothesis, assuming the underlying distribution is symmetric.

The test statistic is

$$t = \frac{\overline{x} - m_0}{s/\sqrt{n}} = 1.2972 < t_{0.10}(14) = 1.3450,$$

so we fail to reject H_0 at 10%.

3. Suppose that $X_1, X_2, \ldots, X_n$ is a i.i.d. random sample from the uniform distribution on $(0, \gamma)$ for some parameter $\gamma > 0$. Assume you don't know γ and you want to estimate it from the sample. Let $X_{(n)}$ be the maximum of $X_1, X_2, \ldots, X_n$.

(a) Show that for $x \in [0, 1]$,

$$P\left(\frac{X_{(n)}}{\gamma} \leq x\right) = x^n.$$

The maximum of $X_1, \ldots, X_n$ is small than y if and only if $X_1 \leq y, \ldots, X_n \leq y$,

$$\begin{aligned} P(X_{(n)} \leq y) &= P(X_1 \leq y, \ldots, X_n \leq y) \\ &= P(X_1 \leq y)P(X_2 \leq y)\ldots P(X_n \leq y) \text{ by independence} \\ &= (y/\gamma)^n \end{aligned}$$

Thus, $P(X_{(n)}/\gamma \leq x) = P(X_{(n)} \leq x\gamma) = x^n$.

(b) Hence show that a $100(1-\alpha)\%$ confidence interval for γ is

$$\left[X_{(n)}, \frac{X_{(n)}}{\alpha^{1/n}}\right].$$

From part(a) we know that

$$P(\frac{X_{(n)}}{\gamma} \leq \alpha^{1/n}) = \alpha,$$

which means

$$P(\gamma \geq X_{(n)}/\alpha^{1/n}) = \alpha$$

which gives the confidence interval above.

(c) Notice that $E(X_1) = \gamma/2$. Using the central limit theorem, show that another $100(1-\alpha)\%$ approximate confidence interval for γ is

$$2\overline{X} \pm \frac{2z_{\alpha/2}S}{\sqrt{n}},$$

where $\overline{X}$ and S^2 are the sample mean and sample variance of $X_1, \ldots, X_n$. Using central limit theorem, we have

$$P\left(-z_{\alpha/2} \leq \frac{\overline{X} - \frac{\gamma}{2}}{S/\sqrt{n}} \leq z_{\alpha/2}\right) = 1 - \alpha$$

$$P\left(2\overline{X} - \frac{2z_{\alpha/2}S}{\sqrt{n}} \leq \gamma \leq 2\overline{X} + \frac{2z_{\alpha/2}S}{\sqrt{n}}\right) = 1 - \alpha.$$

4. (Does Microsoft stock price really follow a lognormal distribution?) In finance literature, one popular model for stock price is the lognormal distribution. Suppose that we have the daily stock price of a stock, say $S_1, S_2, \ldots, S_n$. We can then compute the so-called log-return of the stock price $R_1, R_2, \ldots, R_{n-1}$, where

$$R_i = \ln\left(\frac{S_{i+1}}{S_i}\right), \quad i = 1, \ldots, n-1.$$

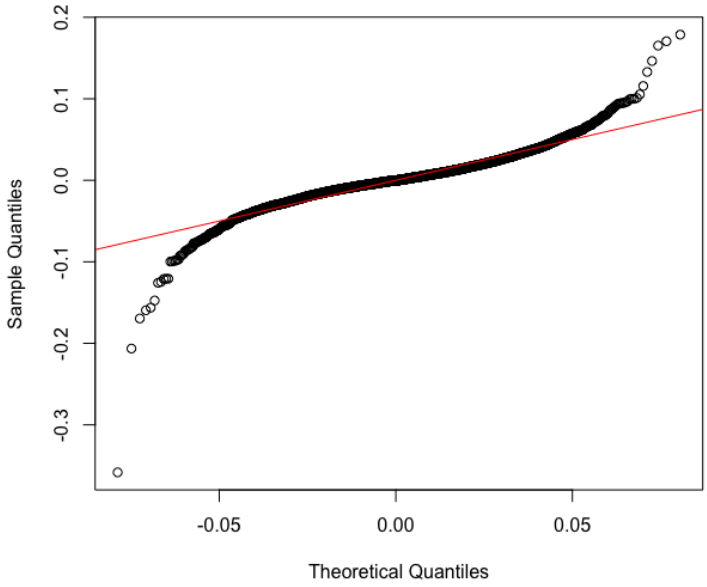
On Blackboard, there is a dataset "MSFT_stock.csv" that contains the daily stock price of Microsoft from March 13, 1986 to October 12, 2020, with $n = 8718$. Use your favourite computing language to answer this question, and attach your code at the end of your assignment.

(a) Compute r_i of the dataset, the realizations of R_i for $i = 1, \ldots, 8717$. Report the sample mean $\overline{r}$ and the sample variance s_r^2.

$$\overline{r} = 0.000385, \quad s_r^2 = 0.000088.$$

(b) Draw a QQ plot of the sample quantiles of r_i against the theoretical quantiles of $N(\overline{r}, s_r^2)$. Label the axis on your plot clearly. **Do not use any existing packages or functions in your computing language that automatically draw the QQ plot for you. Instead, you can use any other functions or packages to determine the sample quantiles and the theoretical quantiles.**

```
#sample code
library(readxl)
data4 <- read_excel('MSFT_stock_2020.xlsx')
Si <-data4$Close
Ri <- log(Si[-1]/Si[-length(Si)])
n <- length(Ri)
plot(qnorm((1:n)/(n+1), mean = mean(Ri), sd = sd(Ri)), sort(Ri),
     xlab = "Theoretical Quantiles", ylab = "Sample Quantiles")
abline(0, 1, col = "red")
```



Note: when computing the theoretical quantiles, remember to specify the mean and standard deviation.

(c) Comment on the QQ plot. Do r_i really follow a normal distribution based on your plot?
The QQ plot does not fall on the line of $y = x$, so it is quite unlikely that r_i follow a normal distribution.

5. (Microsoft stock price continued) In this problem, we will continue to consider the log-return of the stock price $R_1, R_2, \ldots, R_{n-1}$, which are defined in the previous problem. These log-returns are calculated using the formula:

$$R_i = \ln\left(\frac{S_{i+1}}{S_i}\right), \quad i = 1, \ldots, n-1.$$

Denote the median of R_i by m. Use the dataset "MSFT_stock.csv" and your favourite computing language to answer this question, and attach your code at the end of your assignment. **Do not use any existing packages or functions in your computing language that automatically conduct the tests for you (except for (b)). Instead, you can use any other functions or packages to help you conduct these tests.**

(a) From the QQ plot in Assignment 3, it seems that the log-return r_i is symmetric about the origin. Test $H_0 : m = 0$ against $H_1 : m > 0$ using the sign test at $\alpha = 5\%$.
Let n^+ be the number of data that is > 0. Then $n^+ = 4391$, $n = 8717$. The p-value is $P(N^+ \geq 4391; H_0) = 1 - P(Bin(8717, 0.5) \leq 4391) = 1 - 0.7602 = 0.2398 > 0.05$, so we fail to reject H_0 at 5%.

(b) Use R to conduct the Wilcoxon sign-rank test to test the hypothesis in part (a).

```
> wilcox.test(Ri, mu = 0, alternative=c('greater') ,exact=FALSE)

        Wilcoxon signed rank test with continuity correction

data:  Ri
V = 19085294, p-value = 7.277e-06
alternative hypothesis: true location is greater than 0
```

Supplementary:
The Wilcoxon test statistic is $w = 1963637$. The effective sample size, after deleting $r_i = 0$, is $n = 8509$. The p-value is

$$P(W \geq 1963637) = 1 - P(W \leq 1963635) = 1 - \Phi\left(\frac{1963636}{\sqrt{(8509)(8510)(17019)/6}}\right) < 0.05,$$

so we reject H_0 at 5%.

(c) Use the t-test to test the hypothesis in part (a).
The test statistic is

$$t = \frac{\overline{x} - m_0}{s/\sqrt{n}} = 3.8295 > t_{0.05}(8716) \approx z_{0.05} = 1.645,$$

so we reject H_0 at 5%.